

Student: Breno Lopes Mafra

Student ID: 101485572

COMP 3133 – Lab Exercise 2 – Screenshots

- My server.js:

```
- const express = require('express');
- const app = express();
- const PORT = 3000;
-
- app.use(express.static('public'));
-
- app.get('/', (req, res) => {
-   res.send('Hello to Socket.IO Server');
- })
-
- const server = app.listen(PORT, () => {
-   console.log(`Server is running on http://localhost:${PORT}`);
- });
-
- const serverIO = require('socket.io')(server);
-
- serverIO.on('connection', (socket) => {
-   console.log(`A new user connected: ${socket.id}`);
-
-   // Listen for messages from clients
-   socket.on('message', (data) => {
-     console.log('Message received:', data);
-
-     // Broadcast the message to all clients
-     //serverIO.emit('message', data);
-
-     // Broadcast to all clients except the sender
-     //socket.broadcast.emit('message', data);
-
-     // Send back to sender as well
-     socket.emit('message', data);
-   });
- })
```

```

- // Group chat functionality
- socket.on('joinRoom', (room) => {
-   socket.join(room);
-   console.log(`User ${socket.id} joined room: ${room}`);
- });
-
- socket.on('roomMessage', ({ room, message }) => {
-   console.log(`Message to room ${room}: ${message}`);
-   //serverIO.to(room).emit('roomMessage', { sender: socket.id, message });
-   socket.broadcast.to(room).emit('roomMessage', { sender: socket.id, message });
- });
-
- // leave room
- socket.on('leaveRoom', (room) => {
-   socket.leave(room);
-   console.log(`User ${socket.id} left room: ${room}`);
- });
-
- // Handle disconnection
- socket.on('disconnect', () => {
-   console.log(`A user disconnected: ${socket.id}`);
- });
- });

```

- My public/chat.html:

```

- <!DOCTYPE html>
- <html lang="en">
- <head>
-   <meta charset="UTF-8">
-   <meta name="viewport" content="width=device-width, initial-scale=1.0">
-   <title>Chat Page</title>
-   <script src="/socket.io/socket.io.js"></script>
- </head>
- <body>
-   <h1>Chat Page</h1>
-

```

```

<h2>Chat Room</h2>
<div>
  <p>This is a simple chat room using Socket.IO.</p>
  <button onclick="joinRoom('gbc')">Join Room</button>
  <button onclick="leaveRoom('gbc')">Leave GBC Room</button>

  <button onclick="joinRoom('fifa')">Join Fifa Room</button>
  <button onclick="leaveRoom('fifa')">Leave Fifa Room</button>
</div>
  <h2 id="socketID">Socket ID:</h2>
  <h2 id="roomName">No room yet Joined</h2>
  <h3>Messages:</h3>
  <div id="messages"></div>
</div>

<label for="groupMessage">Send Group Message:</label>
<input type="text" id="groupMessage" placeholder="Type your message here">
<button onclick="sendGroupMessage()">Send Message</button>

</div>
<script>

  const messagesDiv = document.getElementById('messages');

  // Initialize Socket.IO client
  const socket = io();
  console.log('Socket.IO client initialized');

  // Example event listener for connection
  socket.on('connect', () => {
    console.log('Connected to server with ID:', socket.id);
    socket.emit('message', 'Hello from client!');
    document.getElementById('socketID').innerText = `Socket ID: ${socket.id}`;
  });

  // Example event listener for messages

```

```

- socket.on('message', (data) => {
-   console.log('New message received:', data);
- })
-
- // Example event listener for room messages
- socket.on('roomMessage', (data) => {
-   console.log('New room message received:', data);
-   const messageElement = document.createElement('p');
-   messageElement.textContent = `ID: ${data.sender} - Room: ${data.room} - Message:
-   ${data.message}`;
-   messagesDiv.appendChild(messageElement);
- })
-
- // Example to send group message
- function sendGroupMessage() {
-   const room = document.getElementById('roomName').innerText || 'gbc'; // Default to 'gbc' if no room
-   specified
-
-   const txtMessage = document.getElementById('groupMessage');
-
-   const data = {
-     sender: socket.id,
-     room: room,
-     message: txtMessage.value
-   };
-
-   socket.emit('roomMessage', { room: room, message: txtMessage.value });
-   console.log(`Sent message to room ${room}: ${txtMessage.value}`);
-
-   const messageElement = document.createElement('p');
-   messageElement.innerHTML = `<b>ID: ${data.sender} - Room: ${data.room} - Message:
-   ${data.message}</b>`;
-   messagesDiv.appendChild(messageElement);
-
-   txtMessage.value = "";
- }
-

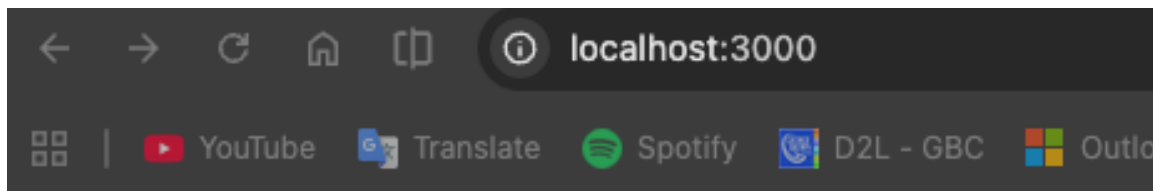
```

```

- // Example event listener for errors
- socket.on('error', (error) => {
-   console.error('Socket.IO error:', error);
- });
-
- // Disconnect event listener
- socket.on('disconnect', () => {
-   console.log('Disconnected from server');
- });
-
- function joinRoom(room) {
-   document.getElementById('roomName').innerText = room;
-   socket.emit('joinRoom', room);
-   console.log(`Joining room: ${room}`);
- }
-
- function leaveRoom(room) {
-   socket.emit('leaveRoom', room);
-   console.log(`Leaving room: ${room}`);
- }
- </script>
- </body>
- </html>

```

- **Server running in localhost/3000:**



Hello to Socket.IO Server

- Test with different clients under different rooms:

localhost:3000/chat.html

Chat Page

Chat Room

This is a simple chat room using Socket.IO.

Socket ID: vjOMuU4ZgD7WyemuAAAB

gbc

Messages:

ID: vjOMuU4ZgD7WyemuAAAB - Room: gbc - Message: Test

ID: vjOMuU4ZgD7WyemuAAAB - Room: fifa - Message: Hi

ID: 9-rlh4ll6oXsTpeAAAD - Room: undefined - Message: Test 2

Send Group Message:

localhost:3000/chat.html

Chat Page

Chat Room

This is a simple chat room using Socket.IO.

Socket ID: 9-rlh4ll6oXsTpeAAAD

gbc

Messages:

ID: 9-rlh4ll6oXsTpeAAAD - Room: gbc - Message: Test 2

ID: 40W6q-SsRnb4sj4qAAAF - Room: undefined - Message: test 4

Send Group Message:

- Console output:

```
Brenos-MacBook-Air:week3 brenolopesmafra$ npm start
> week3@1.0.0 start
> node server.js

Server is running on http://localhost:3000
A new user connected: vjOMuU4ZgD7WyemuAAAB
Message received: Hello from client!
User vjOMuU4ZgD7WyemuAAAB joined room: gbc
Message to room gbc: Test
User vjOMuU4ZgD7WyemuAAAB left room: gbc
User vjOMuU4ZgD7WyemuAAAB joined room: gbc
User vjOMuU4ZgD7WyemuAAAB joined room: fifa
Message to room fifa: Hi
User vjOMuU4ZgD7WyemuAAAB left room: fifa
A new user connected: 9-rlh4ll6oXsTpeAAAD
Message received: Hello from client!
User 9-rlh4ll6oXsTpeAAAD left room: gbc
User 9-rlh4ll6oXsTpeAAAD joined room: fifa
User 9-rlh4ll6oXsTpeAAAD joined room: gbc
Message to room gbc: Test 2
User vjOMuU4ZgD7WyemuAAAB left room: fifa
User vjOMuU4ZgD7WyemuAAAB joined room: gbc
A new user connected: 40W6q-SsRnb4sj4qAAAF
Message received: Hello from client!
Message to room No room yet Joined: test 3
User 40W6q-SsRnb4sj4qAAAF joined room: gbc
User 40W6q-SsRnb4sj4qAAAF left room: gbc
User 40W6q-SsRnb4sj4qAAAF joined room: fifa
User 40W6q-SsRnb4sj4qAAAF left room: fifa
Message to room fifa: test 4
█
```